Summary of the Relevant Properties of Fidelity

We use the **squared-fidelity convention**:

$$F(\rho, \sigma) = \left\| \sqrt{\rho}\sqrt{\sigma} \right\|_1^2,$$

with root fidelity

$$\sqrt{F(\rho, \sigma)} = \left\| \sqrt{\rho}\sqrt{\sigma} \right\|_1.$$

Versus **trace distance**:

$$T(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1,$$

1. Boundedness

$$0 \leq F(\rho, \sigma) \leq 1.$$

Fidelity is a normalized **measure of similarity,** with larger values representing greater overlap.

2. Equality condition

$$F(\rho, \sigma) = 1 \quad \Longleftrightarrow \quad \rho = \sigma.$$

Two normalized quantum states have maximum fidelity exactly when they are the same state.

3. Orthogonality condition

$$F(\rho, \sigma) = 0 \quad \Longleftrightarrow \quad \mathrm{supp}(\rho) \perp \mathrm{supp}(\sigma).$$

Fidelity vanishes exactly when the states have orthogonal supports* and are therefore perfectly distinguishable.

$^*\mathrm{supp}(\rho) = (\ker \rho)^\perp$ is the linear subspace of the domain of ρ that maps to non-zero vectors. It's dimension is the rank of ρ.

4. Symmetry

$$F(\rho, \sigma) = F(\sigma, \rho).$$

Fidelity does not depend on the order in which the two states are compared. fidelity_and_gentle_measurement...

5. ✅ Pure-state formula

For pure states,

$$F(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = |\langle\psi|\phi\rangle|^2.$$

Fidelity reduces to the ordinary squared overlap, or transition probability, between the two state vectors.

6. ✅ Pure–mixed formula

If one state is pure, then

$$F(|\psi\rangle\langle\psi|, \rho) = \langle\psi|\rho|\psi\rangle.$$

This is the probability that ρ passes the projective test for being the target state $|\psi\rangle$.

7. ✅ Commuting or classical case

If

$$\rho = \sum_x p_x |x\rangle\langle x|, \qquad \sigma = \sum_x q_x |x\rangle\langle x|,$$

then

$$F(\rho, \sigma) = \left(\sum_x \sqrt{p_x q_x} \right)^2.$$

Quantum fidelity reduces to the square of the classical Bhattacharyya overlap when the states commute. fidelity_and_gentle_measurement...

8. 📌 Uhlmann's theorem

$$F(\rho, \sigma) = \max_{\substack{|\psi_\rho\rangle,\, |\psi_\sigma\rangle \\ \text{purifications of } \rho, \sigma}} |\langle\psi_\rho|\psi_\sigma\rangle|^2.$$

Fidelity is the largest possible squared overlap between purifications of the two mixed states. fidelity_and_gentle_measurement...

9. Unitary and isometric invariance

For every unitary or isometry V,

$$F\big(V\rho V^\dagger, V\sigma V^\dagger\big) = F(\rho, \sigma).$$

Reversible embeddings and unitary evolution preserve fidelity exactly. talk_3_outline(8)

10. Multiplicativity under tensor products

$$F(\rho_1 \otimes \rho_2, \sigma_1 \otimes \sigma_2) = F(\rho_1, \sigma_1)\, F(\rho_2, \sigma_2).$$

The fidelity of product states is the product of the fidelities of their individual components. fidelity_and_gentle_measurement…

11. 📌 Monotonicity under quantum channels

For every quantum channel $\mathcal{N}$,

$$F(\mathcal{N}(\rho), \mathcal{N}(\sigma)) \geq F(\rho, \sigma).$$

Applying the same physical process to both states cannot decrease their fidelity; noise can make states more similar. fidelity_and_gentle_measurement…

12. Monotonicity under partial trace

For bipartite states,

$$F(\rho_A, \sigma_A) \geq F(\rho_{AB}, \sigma_{AB}),$$

where

$$\rho_A = \mathrm{Tr}_B\, \rho_{AB}, \qquad \sigma_A = \mathrm{Tr}_B\, \sigma_{AB}.$$

Discarding a subsystem cannot decrease fidelity because it removes potentially distinguishing information.

13. Behaviour under measurements

For a POVM $\{\Lambda_y\}_y$, define

$$p_y = \mathrm{Tr}(\Lambda_y \rho), \qquad q_y = \mathrm{Tr}(\Lambda_y \sigma).$$

Then

$$\sum_y \sqrt{p_y q_y} \geq \sqrt{F(\rho, \sigma)}.$$

Every measurement produces classical distributions whose overlap is at least the root fidelity of the original quantum states. talk_3_outline(8)

14. ‼️ Fuchs–van de Graaf inequalities

With trace distance

$$T(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1,$$

we have

$$1 - \sqrt{F(\rho, \sigma)} \leq T(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)}.$$

Fidelity and trace distance define compatible notions of closeness and allow estimates in one measure to be converted into estimates in the other. fidelity_and_gentle_measurement...

15. 📌 Exact pure-state relation with trace distance

For pure states,

$$T(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = \sqrt{1 - F(\psi, \phi)}.$$

For pure states, fidelity and trace distance determine one another exactly.

16. 📌 Purified distance

$$P(\rho, \sigma) = \sqrt{1 - F(\rho, \sigma)}.$$

Purified distance converts fidelity into a genuine distance that behaves especially well with purifications and approximate quantum states.

17. 📌 Contraction of purified distance under channels

$$P(\mathcal{N}(\rho), \mathcal{N}(\sigma)) \leq P(\rho, \sigma).$$

Physical processing cannot increase purified distance, just as it cannot increase trace distance. fidelity_and_gentle_measurement...

18. Fidelity itself is not a metric

Fidelity is a similarity measure rather than a distance: identical states have fidelity 1, and fidelity itself does not satisfy the usual metric axioms such as the triangle inequality.